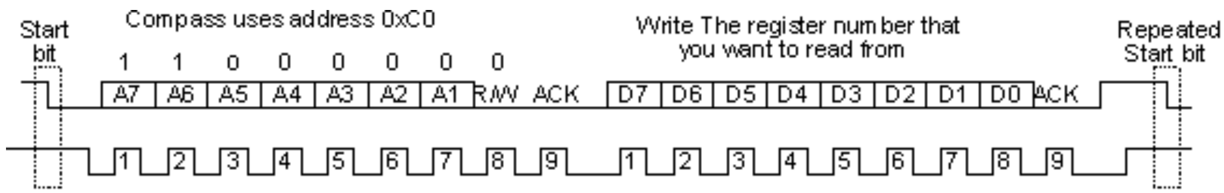
I2C Writing Master

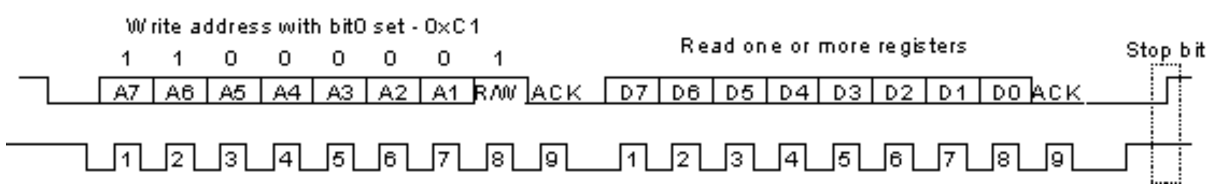
The I2C protocol is a synchronous, multi-master/multi-slave (controller/target), packet-switched, single-ended, serial communication bus. It is widely used for attaching lower-speed peripheral ICs to processors and microcontrollers in short-distance, intra-board communication. The bus is composed of two wires: SDA, and SCL. The former is the data line, while the latter is the clock line. To address different slaves connected on the shared bus, the protocol uses a 7-bit addressing scheme (up to 128 slaves) to transmit 8-bit data packets.

The communication works as here described:

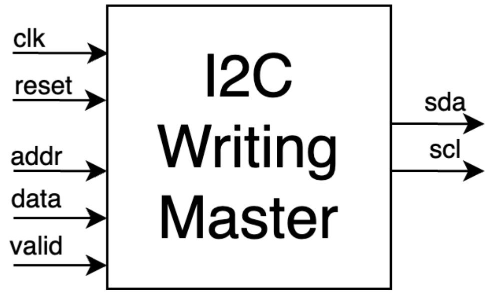
1. The master starts the transmission by sending a **START** bit followed by the 7-bit address of the slave it wishes to communicate with, which is finally followed by a single bit representing whether it wishes to write (0) to or read (1) from the slave.
2. If the slave exists on the bus then it will respond with an **ACK** bit (active low for acknowledged) for that address. The master then continues in either transmit or receive mode (according to the read/write bit it sent), and the target continues in the complementary mode (receive or transmit, respectively).
3. The address and the data bytes are sent with the most significant bit first. The **START** condition is indicated by a high-to-low transition of SDA with SCL high; the **STOP** condition is indicated by a low-to-high transition of SDA with SCL high. All other transitions of SDA take place with SCL low.
4. If the master wishes to write to the slave, then it repeatedly sends a byte with the slave sending an ACK bit. (In this situation, the master is in transmit mode, and the slave is in receive mode.)
5. If the master wishes to read from the slave, then it repeatedly receives a byte from the slave, while sending an ACK bit after every byte except the last one. (In this situation, the master is in receive mode, and the slave is in transmit mode.)
6. An I^2C transaction may consist of multiple messages. The master terminates a message with a **STOP** condition if this is the end of the transaction or it may send another **START** condition to retain control of the bus for another message (a "combined format" transaction).

These are two example transactions:





It is required to design a digital circuit that realises **JUST** the writing part of the protocol. The interface of the circuit to be designed is as follows:



For simplicity, the SCL clock is 32 times slower than the system clock. While idle, the circuit will wait for the valid signal to be asserted in order to start the writing transaction with the provided data at the indicated slave address.

The slave acknowledgements should be simulated in the Testbench.

You are requested to deal with the various possible error situations, documenting the choices made.

The final project report must contain:

- Introduction (circuit description, possible applications, possible architectures, etc.)
- Description of the architecture designed (block diagram, inputs/outputs, etc.)
- VHDL code (with detailed comments) to be attached to the report.
- Test strategy (Test-plan) and related Testbench for verification; a detailed, though not exhaustive, verification is required, including error situations and borderline cases of functioning
- Interpretation of the results obtained in the automatic synthesis/implementation on a Xilinx FPGA platform in terms of maximum clock frequency (critical path), elements used (slice, LUT, etc.) and estimated power consumption. Comment on any warning messages.
- Conclusions